Advanced Machine Learning, ETH Zurich

# Hinge-fan networks

*One hidden layer, one hinge per kink*

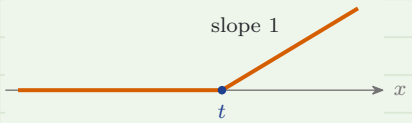
Carlos Cotrini

Autumn semester 2026

Entry 1

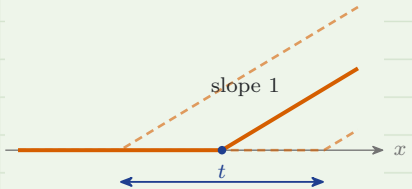
# One layer of hinges

A hinge is the ReLU of a shifted input, $\mathrm{ReLU}(x - t)$: it stays at 0 left of $t$ and rises with slope 1 right of it. Its position $t$ is the only number it has.



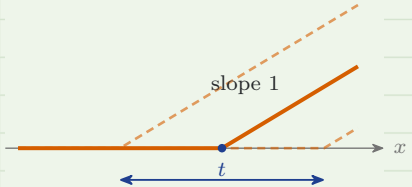

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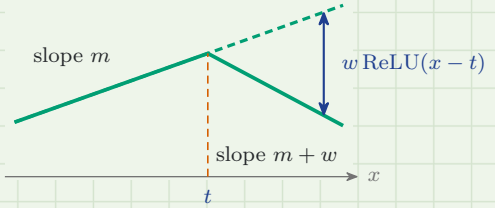



Solid: the hinge $\mathrm{ReLU}(x - t)$. Dashed: the same hinge at two other positions. Changing $t$ slides it along $x$ and keeps its shape.

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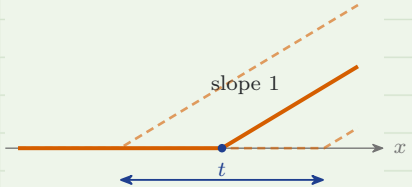



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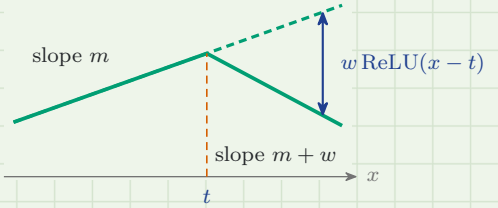



Add $w\,\mathrm{ReLU}(x - t)$ to a curve of slope $m$ (dashed): from $t$ on, its slope is $m + w$ (solid).

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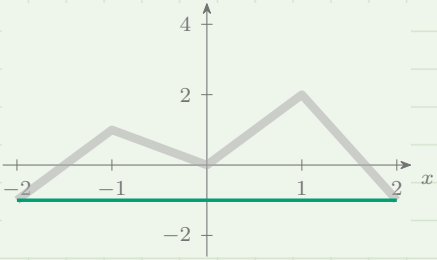



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The weight $w$ turns the curve at $t$, up for $w > 0$ and down for $w < 0$. In a network, $t$ sits in the hinge and $w$ on the arrow that leaves it.

Start from the constant $c$. A hinge at the left end $a$ is a line on $[a, b]$, so it sets the first slope; every further hinge turns the curve at its position $t_j$ by its weight $w_j$.

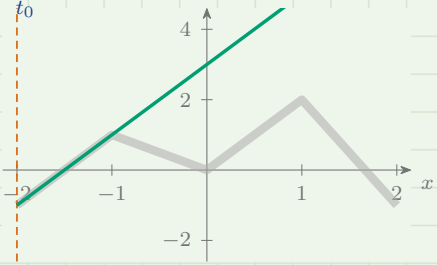
$$y = c + \sum_j w_j \,\mathrm{ReLU}(x - t_j)$$




Grey: the zigzag $f$. Green: the sum so far, one hinge more per click.

$y = -1$ the bias $c = f(-2)$

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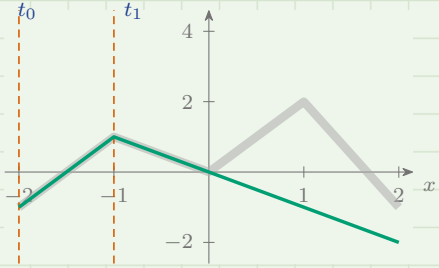
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$+\,2\,\mathrm{ReLU}(x+2)$ $t_0 = -2$, slope 0 to 2

On $[-2, 2]$, $\mathrm{ReLU}(x+2) = x+2$: hinge 0 sits at the left end and is a line, so with $c$ it gives the first piece, $2x + 3$.

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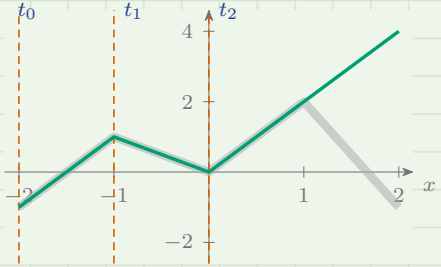

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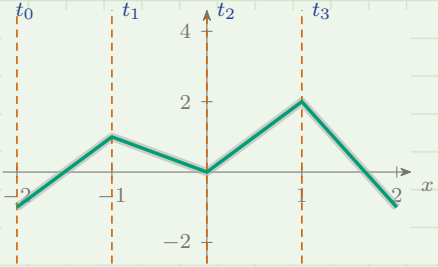

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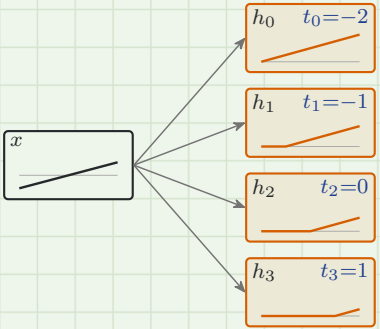

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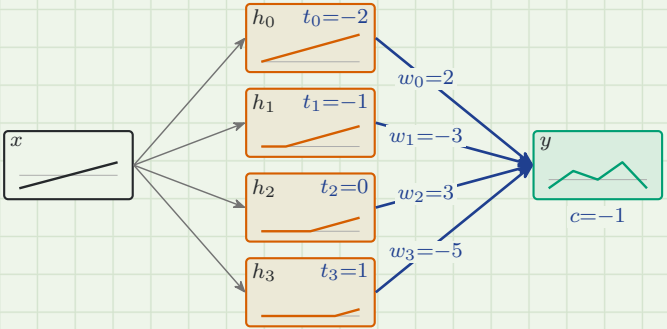
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Hinge $j$ reads $x$ with the fixed weight 1 and bends at its position $t_j$. The output $y$ adds the hinges up, with the weights $w_j$ and the bias $c$.




Every box draws its neuron on $[-2, 2]$. The hinges, on one scale, are one ramp slid to $-2$, $-1$, 0, 1.

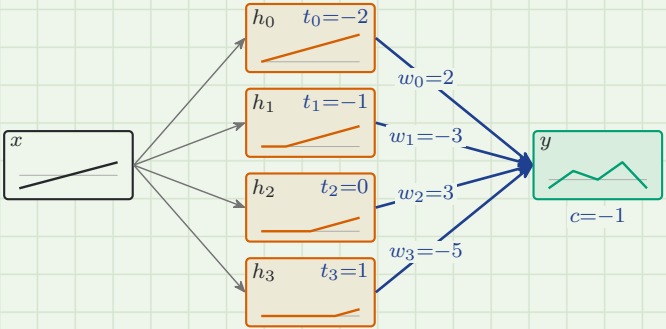
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$$h_j = \mathrm{ReLU}(x - t_j)$$

$$y = c + \textstyle\sum_j w_j h_j$$

Free, in pen: $t_j$ and $w_j$ for every hinge, and $c$: $2n + 1$ numbers, nine here. A grey arrow has the weight 1. The hinges can come in any order.

Entry 2

# The construction

Read the numbers off $f$: the bias is its value at $a$, hinge 0 sits at $a$ with the first slope $m_0$, and hinge $i$ sits at the kink $t_i$ with the change of slope $\Delta_i = m_i - m_{i-1}$ there.

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|---|---|---|---|
| output | $c = f(a)$ | | |
| hinge 0 | $t_0 = a$, | $w_0 = m_0$ | |
| hinge $i$ | $t_i$, its kink, | $w_i = \Delta_i$ | $i = 1, \dots, K$ |



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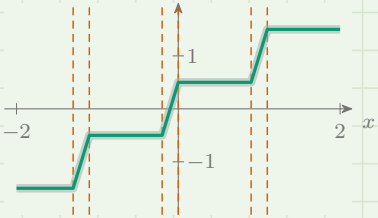
Then $y = f(a) + m_0 \,\mathrm{ReLU}(x - a) + \sum_i \Delta_i \,\mathrm{ReLU}(x - t_i)$,
and $y = f$: both are piecewise linear, they agree at $a$,
and on the $i$th piece both have the slope
$m_0 + \Delta_1 + \cdots + \Delta_i = m_i$.

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$K$ kinks take $K + 1$ hinges, and $K$ when $f$ starts flat: then $m_0 = 0$ and hinge 0 goes.




The staircase of the page starts flat: six kinks, six hinges, $c = -1.5$ and $w_j = 5, -5, 5, -5, 5, -5$.

A hidden neuron usually has two numbers, $\mathrm{ReLU}(ux+v)$, and bends at $t=-v/u$. On $[a,b]$ each such neuron is a hinge of ours with a weight, plus at most a line and a constant.

$u>0$: $\mathrm{ReLU}(ux+v)=u\,\mathrm{ReLU}(x-t)$.
$u$ moves onto the outgoing weight:
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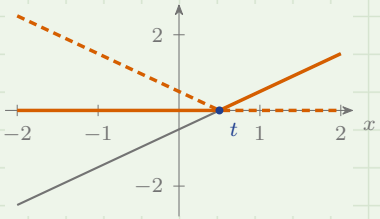
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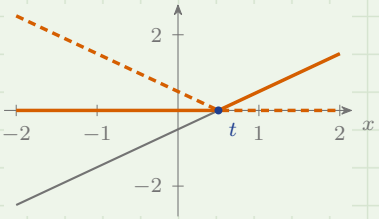

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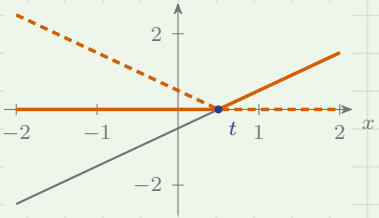
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$u=0$: a constant, which $c$ takes.

So $n$ arbitrary neurons make at most $n+1$ of our hinges: the one at $a$ collects every line.




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Entry 3

# Fitting

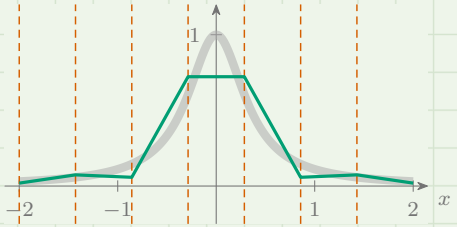
Fix the positions $t_1, \dots, t_n$. Then $y = c + \sum_j w_j \varphi_j(x)$, with the features $\varphi_j(x) = \mathrm{ReLU}(x - t_j)$, is linear in the numbers $\theta = (c, w_1, \dots, w_n)$ that are left.

On the points $x_1, \dots, x_N$, with $f$ the target's values there, the mean squared error $\frac{1}{N}\|\Phi\theta - f\|^2$ is a convex quadratic in $\theta$. Row $i$ of the matrix $\Phi$ holds the features at $x_i$:

$$\big(1,\ \varphi_1(x_i),\ \dots,\ \varphi_n(x_i)\big).$$

Its minimizers solve the normal equations

$$\Phi^\top \Phi\, \theta = \Phi^\top f$$




Seven hinges spread evenly from $-2$ straddle the peak, and the best weights miss it: MSE $8.0 \times 10^{-3}$.

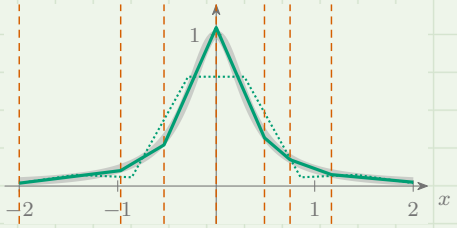
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Placed by the fit of the next frame, one hinge sits on the peak: MSE $1.3 \times 10^{-4}$, about sixty times lower. Dotted: the even fit.

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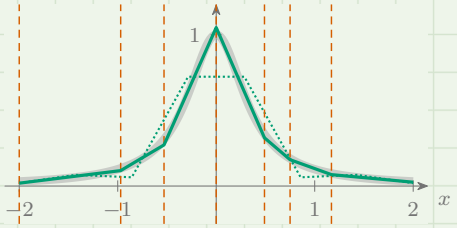
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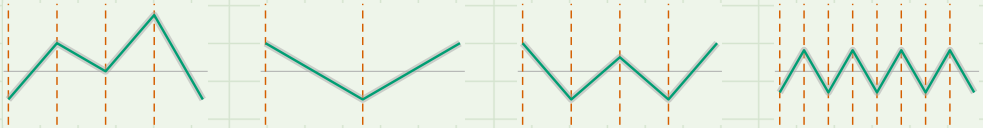
The hidden layer picks the features, and the output is linear regression on them, as in the course; random features pick the hidden layer at random. Placing the hinges is the nonconvex part.




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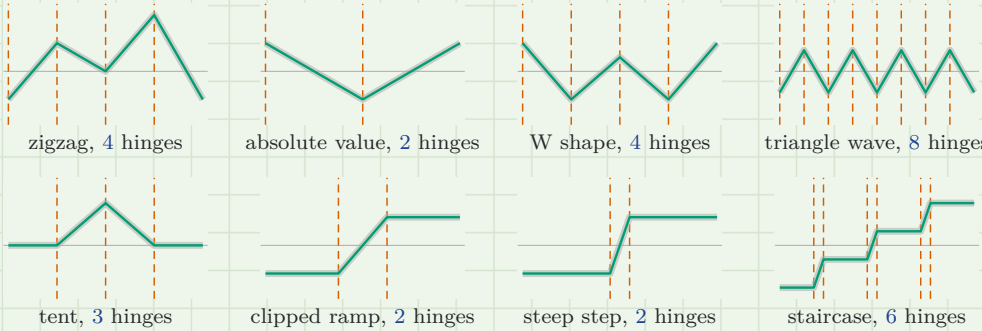
Placed by the fit of the next frame, one hinge sits on the peak: MSE $1.3 \times 10^{-4}$, about sixty times lower. Dotted: the even fit.

The page's fit places the hinges by a dynamic program, with a hinge at $a$ and without, sets $c$ and the $w_j$ by least squares, and keeps the smaller error.



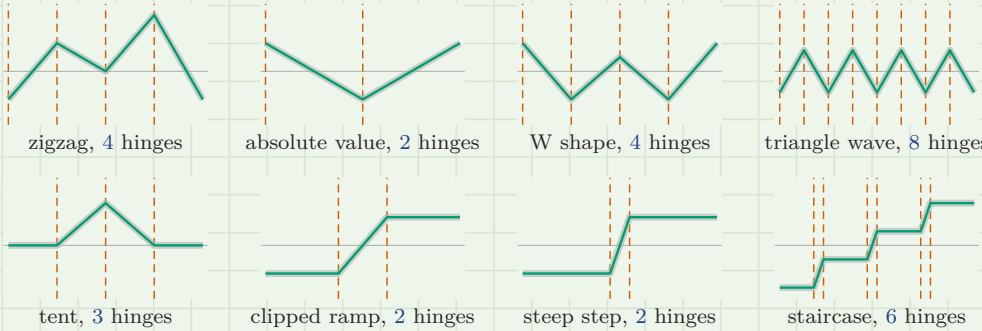

Grey: the target, under the fit; dashed: the hinges. Top row: one hinge per kink, and one for the first slope.

The page's fit places the hinges by a dynamic program, with a hinge at $a$ and without, sets $c$ and the $w_j$ by least squares, and keeps the smaller error.



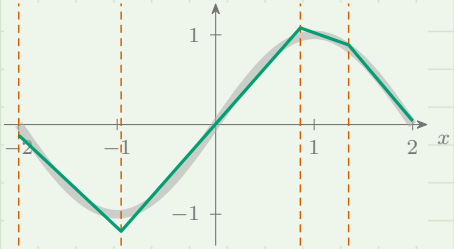

Grey: the target, under the fit; dashed: the hinges. Top row: one hinge per kink, and one for the first slope. Bottom row: these start flat, so one per kink is enough.

The page's fit places the hinges by a dynamic program, with a hinge at $a$ and without, sets $c$ and the $w_j$ by least squares, and keeps the smaller error.




Grey: the target, under the fit; dashed: the hinges. Top row: one hinge per kink, and one for the first slope. Bottom row: these start flat, so one per kink is enough. No fit with one hinge fewer is exact.

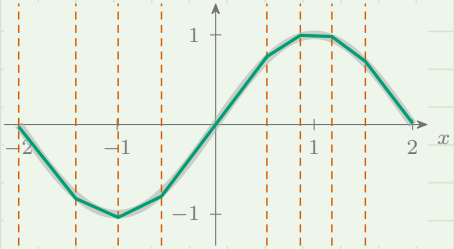
If $|f''| \le M$, the interpolant through $n+1$ evenly spaced points is a network of $n$ hinges, and it stays within $2M/n^2$ of $f$ on $[-2, 2]$.




The sine $\sin(\pi x/2)$ with 4 hinges: MSE $3.7 \times 10^{-3}$, largest error 0.19.

Each doubling of the hinges divides the largest error by about 4, and the MSE by about 16.

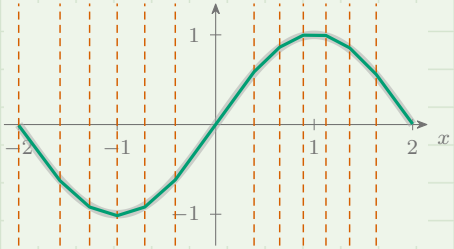
If $|f''| \le M$, the interpolant through $n+1$ evenly spaced points is a network of $n$ hinges, and it stays within $2M/n^2$ of $f$ on $[-2, 2]$.




The sine $\sin(\pi x/2)$ with 8 hinges: MSE $1.7 \times 10^{-4}$, largest error 0.040.

Each doubling of the hinges divides the largest error by about 4, and the MSE by about 16.

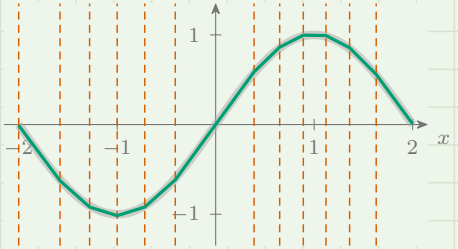
If $|f''| \le M$, the interpolant through $n+1$ evenly spaced points is a network of $n$ hinges, and it stays within $2M/n^2$ of $f$ on $[-2, 2]$.




The sine $\sin(\pi x/2)$ with 12 hinges: MSE $3.3 \times 10^{-5}$, largest error 0.017.

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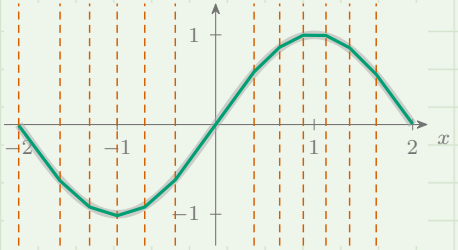

The sine $\sin(\pi x/2)$ with 12 hinges: MSE $3.3 \times 10^{-5}$, largest error 0.017.

Each doubling of the hinges divides the largest error by about 4, and the MSE by about 16.

Hinges until the largest error drops below 0.02, about two pixels on the page:

| | | | |
|---|---|---|---|
| parabola | 9 | sine | 11 |
| Runge function | 9 | cubic | 11 |
| Gaussian bump | 11 | two periods | 21 |
| | | chirp | 24 |

If $|f''| \le M$, the interpolant through $n+1$ evenly spaced points is a network of $n$ hinges, and it stays within $2M/n^2$ of $f$ on $[-2, 2]$.




The sine $\sin(\pi x/2)$ with 12 hinges: MSE $3.3 \times 10^{-5}$, largest error 0.017.

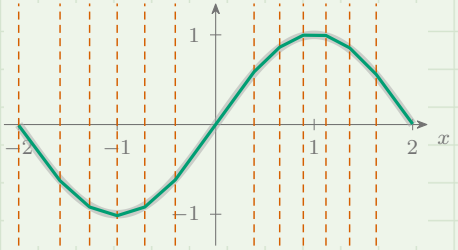
Each doubling of the hinges divides the largest error by about 4, and the MSE by about 16.

Hinges until the largest error drops below 0.02, about two pixels on the page:

| | | | |
|---|---|---|---|
| parabola | 9 | sine | 11 |
| Runge function | 9 | cubic | 11 |
| Gaussian bump | 11 | two periods | 21 |
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Least squares beats interpolating $f$ at the same kinks: with $n = 10$ it lowers the MSE about 4 to 6 times, for the sine to $7.0 \times 10^{-5}$ from $3.6 \times 10^{-4}$.

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Try them on the page: eth-aml-hs26.github.io/neural-networks/viz/hinge-fan-networks

Entry 4

# Summary

The hinge chain and the hinge fan both build a piecewise-linear function out of hinges. They differ in depth, in width, and in what each hinge reads.

| | **Hinge chain** | **Hinge fan** |
|---|---|---|
| Hidden layers | $K$ | 1 |
| Neurons per layer | 2 | $K+1$ ($K$ if $f$ starts flat) |
| A hinge reads | the hinge before it | $x$ |

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| A hinge reads | the hinge before it | $x$ |
| ReLU on | every neuron, lift $C$ | the hinges |
| Free numbers | gaps $d_\ell$, slope changes $\Delta_\ell$ | $t_j, w_j$; $c$ |
| The kinks | in order, by the gaps | anywhere |

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Each network has its page, where you pick a target and set every free number by hand or let fit set them:

*Hinge-fan networks* `eth-aml-hs26.github.io/neural-networks/viz/hinge-fan-networks`
*Hinge-chain networks* `eth-aml-hs26.github.io/neural-networks/viz/hinge-chain-networks`